

Paper 18.50 - VOA / Moonshine Route Claim Contract

CQE-paper-18.50

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-18.50.md

Paper 18.50 - VOA / Moonshine Route Claim Contract

Purpose

This contract governs all Paper 18 claims. It admits finite seed and bounded route receipts while rejecting global Moonshine or extractor promotions.

Admitted Claims

Admit a claim when it gives:

```
route source
route target
finite or bounded scope
verifier status
honesty label
open route residue
```

Rejected Promotions

Reject by default:

```
finite VOA seed -> full Moonshine identification
bounded McKay table -> full McKay-Thompson arithmetic
registered correction class -> proved Rule 30 extractor
Monster scalar -> physical representation theorem
pass_with_open_gaps -> closed theorem
```

Hidden-Guess Rule

All route diagnostics must hide the receipt until the reviewer has classified the route as:

```
closed finite
bounded table
bounded open-gap
open
invalid
```

Linked Receipt

Current receipt:

```
production/formal-papers/CQE-paper-18/voa_moonshine_routes_receipt.json
```

Current verdict:

```
finite VOA seed: pass
static Z4 route template: pass
bounded McKay bootstrap: pass
correction_via_voa: open
full Moonshine route: open
```

Boundary Failure

The contract fails if a later paper says Paper 18 completes `correction\_via\_voa`, proves the global Rule 30 extractor, or identifies the finite seed with full Moonshine without a transport theorem.

It passes if a later paper uses Paper 18 as bounded route evidence and carries the open global residues forward.

Conclusion

Paper 18.50 keeps bounded route evidence useful without letting it become a hidden global proof claim.